9

Disorders of Water Metabolism

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PHYSIOLOGY OF WATER BALANCE

The maintenance of the tonicity of body fluids within a narrow physiologic range is made possible by homeostatic mechanisms that control the intake and excretion of water. Vasopressin, known as arginine vasopressin (AVP) or antidiuretic hormone (ADH), governs the excretion of water by its effect on the renal collecting system. Osmoreceptors located in the hypothalamus control the secretion of vasopressin in response to changes in tonicity.

In the steady state, water intake matches water losses. Water intake is regulated by the need to maintain a physiologic serum osmolality of 285 to 290 mOsm/kg H_2O. Despite major fluctuations of solute and water intake, the total solute concentration (i.e., the tonicity) of body fluids is maintained virtually constant. The ability to dilute and to concentrate the urine allows wide flexibility in urine flow (see Chapter 2). During water loading, the diluting mechanisms permit excretion of 20 to 25 L of urine daily, and during water deprivation, the urine volume may be as low as 0.5 L/day.[1–3]

VASOPRESSIN

AVP plays a critical role in determining the concentration of urine: It is a 9-amino acid cyclic peptide, synthesized and secreted by the supraoptic and paraventricular magnocellular nuclei in the hypothalamus. AVP has a half-life of 15 to 20 minutes and is rapidly metabolized in the liver and the kidney.

Osmotic Stimuli for Vasopressin Release

Substances restricted to the extracellular fluid (ECF), such as hypertonic saline and mannitol, decrease cell volume by acting as effective osmoles and enhancing osmotic water movement from the cell. This stimulates vasopressin release; in contrast, urea and glucose readily cross cell membranes and thus do not cause changes in cell volume. The "osmoreceptor" cells, located close to the supraoptic nuclei in the anterior hypothalamus, are sensitive to changes in serum osmolality as small as 1% and bring about the release of AVP by a pathway that involves the activation of transient receptor potential (TRPV4) channels.[4] In humans, the osmotic threshold for AVP release is 280 to 290 mOsm/kg H_2O[5] (Fig. 9.1). This system is so efficient that serum osmolality usually does not vary by more than 1% to 2% despite wide fluctuations in water intake.

Nonosmotic Stimuli for Vasopressin Release

Decreased effective circulating volume (e.g., heart failure, cirrhosis, vomiting) causes discharge from parasympathetic afferent nerves in the carotid sinus baroreceptors and increases AVP secretion. Much higher vasopressin levels can be achieved with hypovolemia than with hyperosmolality, although a large (7%) decrease in blood volume is required before this response is elicited. Other nonosmotic stimuli include nausea, pain, pregnancy, and fructose (such as from sugary beverages).[6] AVP levels (as reflected by elevation of serum copeptin, a surrogate marker) are also elevated in patients with metabolic syndrome and in patients with chronic kidney disease (CKD) compared with healthy individuals. The function of the V1a and V1b receptors in human disease are still being evaluated, although a recent study suggests that V1b may have a role in driving fat in response to fructose, possibly as a means for increasing metabolic water reserves.[6]

Mechanism of Vasopressin Action

AVP binds three types of receptors coupled to G proteins: the V_{1a} (vascular and hepatic), V_{1b} (anterior pituitary and pancreatic islet), and V_2 (renal) receptors. The V_2 receptor is primarily localized in the collecting duct and leads to an increase in water permeability through aquaporin 2 (AQP2), a member of a family of cellular water transporters (Fig. 9.2). AQP1 is localized in the apical and basolateral region of the proximal tubule epithelial cells and the descending limb of the loop of Henle and accounts for the high water permeability of these nephron segments. AQP1 is constitutively expressed, and not subject to regulation by AVP.[7] AQP2 is localized exclusively in apical plasma membranes and intracellular vesicles in the collecting duct principal cells. AVP affects both the short-term and the long-term regulation of AQP2. The short-term regulation, also described as the *shuttle hypothesis,* explains the rapid and reversible increase (within minutes) in collecting duct water permeability after AVP administration. This involves the insertion of water channels from subapical vesicles into the luminal membrane. Long-term regulation relates to AVP-mediated increases in the transcription of genes involved in AQP2 production and occurs if AVP levels are elevated for 24 hours or more. An increased number of AQP2 channels per cell elevates the maximal water permeability of the collecting duct epithelium.[7]

Aquaporins 3 and 4 are located on the basolateral membranes of the collecting duct (Fig. 9.2) and are probably involved in water exit from the cell. AQP3 is expressed along the entire collecting duct, whereas AQP4 is found in the inner medullary collecting duct. AQP4 is also found in the hypothalamus and is a candidate osmoreceptor for the control of AVP release.[7]

AVP also stimulates urea transporters in the inner medullary collecting duct. Urea transporter A1 (UT-A1) is found exclusively in apical plasma membranes and intracellularly in inner medullary collecting ducts. UT-A3 is found predominantly in the basolateral plasma membrane in inner medullary collecting ducts. Urea reabsorption into the inner medullary interstitium is important for increasing inner medullary tonicity and the driving force for water reabsorption.[8]

Thirst and Water Balance

Hypertonicity is the most potent stimulus for thirst, with a change of only 2% to 3% in serum osmolality producing a strong desire to drink.

The osmotic threshold for thirst usually occurs at 290 to 295 mOsm/kg H_2O and is greater than the threshold for vasopressin release (see Fig. 9.1). This osmolality closely approximates the level at which maximal concentration of urine is achieved. Hypovolemia, hypotension, and angiotensin II (Ang II) are also stimuli for thirst. Between the limits imposed by the osmotic thresholds for thirst and AVP release, serum osmolality may be regulated more precisely by small, osmoregulated adjustments in urine flow and water intake. The exact level at which balance occurs depends on insensible losses, water intake, and water generated from metabolism.

QUANTITATION OF RENAL WATER EXCRETION

Urine volume can be considered as having two components. The osmolar clearance (C_{osm}) is the volume needed to excrete solutes at the concentration of solutes in serum. The free water clearance (C_{water}) is the volume of water that has been added to (positive C_{water}) or subtracted from (negative C_{water}) isotonic urine (C_{osm}) to create either hypotonic or hypertonic urine.

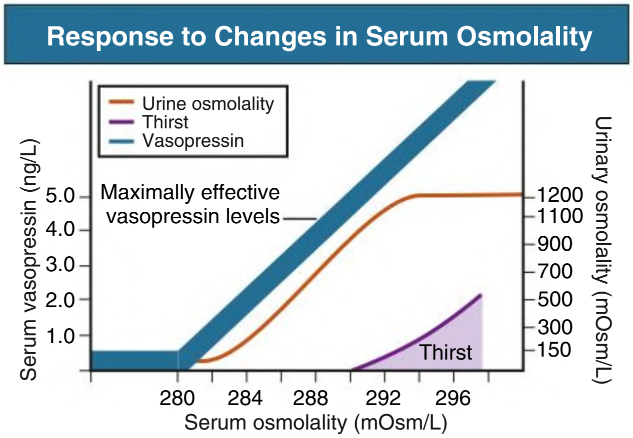

Fig. 9.1 Mechanisms Maintaining Serum Osmolality. Thirst, vasopressin levels, and urinary osmolality in response to changes in serum osmolality. (Modified from Narins RG, Krishna GC. Disorders of water balance. In: Stein JH, ed. *Internal Medicine.* Little, Brown; 1987:794.)

Urine volume flow (V) includes the isotonic portion of urine (C_{osm}) plus the free water clearance (C_{water}).

$$V = C_{osm} + C_{water}$$

Therefore:

$$C_{water} = V - C_{osm}$$

The C_{osm} solute clearance is determined by urine flow, urine osmolality, and serum osmolality P_{osm} as follows:

$$C_{osm} = \left(\frac{U_{osm} \times V}{P_{osm}}\right)$$

Therefore:

$$C_{water} = V - \left(\frac{U_{osm} \times V}{P_{osm}}\right) = V\left(1 - \frac{U_{osm}}{P_{osm}}\right)$$

This relationship reflects the following:

1. In hypotonic urine ($U_{osm} < P_{osm}$), C_{water} is positive.
2. In isotonic urine ($U_{osm} = P_{osm}$), C_{water} is zero.
3. In hypertonic urine ($U_{osm} > P_{osm}$), C_{water} is negative (i.e., water is retained).

If excretion of free water in a polyuric patient is unaccompanied by water intake, the patient becomes hypernatremic. Conversely, failure to excrete free water with increased water intake can cause hyponatremia.

A limitation of the previous equation is that it fails to predict clinically important alterations in serum tonicity and serum sodium concentration (serum $[Na^+]$) because urea is included in the calculation of urine osmolality. Urea is an important component of urinary osmolality; however, urea equilibrates across most cell membranes in the body and does

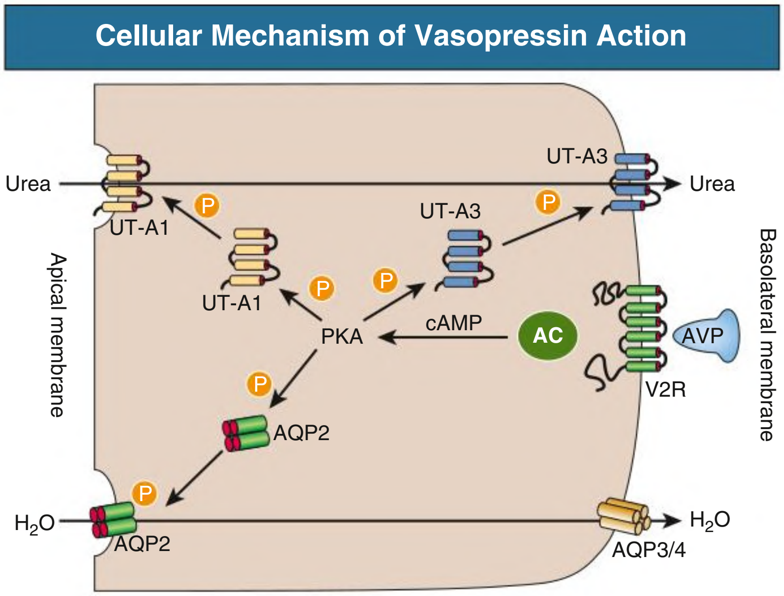

Fig. 9.2 Cellular Mechanism of Vasopressin Action. Arginine vasopressin (AVP) binds to V_2 receptors (V2R) on the basolateral membrane and activates G proteins, stimulates adenylyl cyclase (AC), and initiates a signaling cascade (cAMP, PKA) resulting in aquaporin 2 (AQP2) and urea transporter A1 (UT-A1) phosphorylation (P) and accumulation in the apical membrane. Water exits the cell through AQP3 and AQP4 in the basolateral membrane. Urea exits the cell through urea transporter A3 (UT-A3) in the basolateral membrane. *cAMP,* Cyclic adenosine monophosphate; *PKA,* protein kinase A.

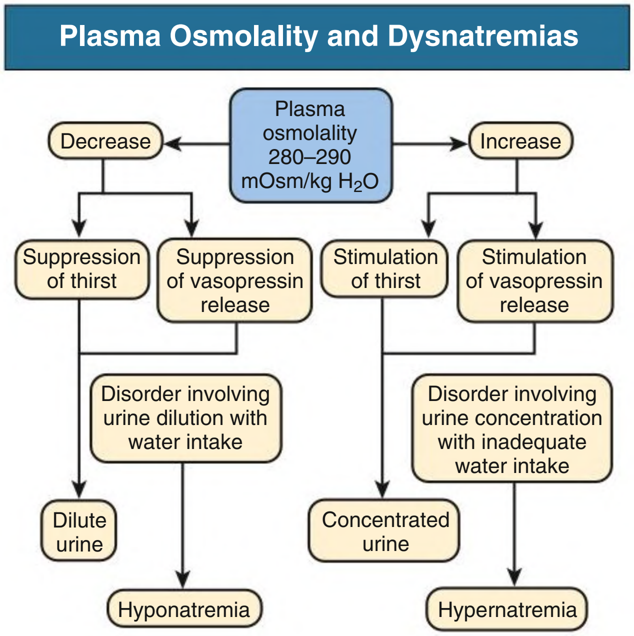

Fig. 9.3 Maintenance of serum osmolality and pathogenesis of dysnatremias. (Modified from Halterman R, Berl T. Therapy of dysnatremic disorders. In: Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension.* Saunders; 1999:257–269.)

not cause water movement between fluid compartments. Therefore, urea does not influence serum Na^+ concentration or the release of vasopressin, and changes in serum $[Na^+]$ are better predicted by electrolyte free water clearance [Cwater(e)]. The equation can be modified, replacing P_{osm} by serum $[Na^+]$ (P_{Na}), and the urine osmolality by urine $[Na^+]$ and urine $[K^+]$, potassium concentration, ($U_{Na} + U_K$):

$$C_{water}(e) = V\left(1 - \frac{U_{Na} + U_K}{P_{Na}}\right)$$

If $U_{Na} + U_K$ is less than P_{Na}, then $C_{water}(e)$ is positive and serum $[Na^+]$ increases. If $U_{Na} + U_K$ is greater than P_{Na}, then $C_{water}(e)$ is negative and serum $[Na^+]$ decreases. In the clinical setting, it is more appropriate to use the equation for electrolyte free clearance to predict if a patient's serum $[Na^+]$ will increase or decrease in the face of the prevailing water excretion. For example, in a patient with high urea excretion, the original equation would predict negative water excretion and a decrease in serum $[Na^+]$, but, in fact, $[Na^+]$ increases, which is accurately predicted by the latter equation.

SERUM SODIUM CONCENTRATION, OSMOLALITY, AND TONICITY

The kidney's countercurrent mechanism, which allows urinary concentration and dilution, acts in concert with the hypothalamic osmoreceptors through AVP secretion to maintain serum $[Na^+]$ and tonicity within a very narrow range[9] (Fig. 9.3). A defect in the urine-diluting capacity coupled with excess water intake leads to hyponatremia. A defect in urine-concentrating ability with inadequate water intake leads to hypernatremia.

Serum $[Na^+]$ along with its accompanying anions accounts for nearly all the osmotic activity of the serum. Calculated serum osmolality is given by $2[Na^+]$ mmol/L + BUN (mg/dL)/2.8 + glucose (mg/dL)/18, where BUN is blood urea nitrogen and 2.8 and 18 represent correction factors based on molecular weights of the respective molecules so as to unify the equation units to mmol/L. The addition of

TABLE 9.1 Effects of Osmotically Active Substances on Serum Sodium (Na^+) Levels

Substances That Increase Osmolality Without Changing Serum Na^+	Substances That Increase Osmolality and Decrease Serum Na^+ (Translocational Hyponatremia)
Urea	Hyperglycemia
Ethanol	Mannitol
Ethylene glycol	Glycine
Isopropyl alcohol	Maltose
Methanol	

other solutes to ECF results in an increase in measured osmolality (Table 9.1). Solutes that are permeable across cell membranes do not cause water movement and therefore yield hypertonicity without cellular dehydration, as in uremia or ethanol intoxication. By contrast, in diabetic ketoacidosis, when glucose cannot freely cross cell membranes in the absence of insulin, water moves from the cells to the ECF, leading to cellular dehydration and lowering serum $[Na^+]$. This can be viewed as translocational because the decrease in serum $[Na^+]$ does not reflect a change in total body water but rather the movement of water from intracellular to extracellular space. Therefore, serum $[Na^+]$ can be corrected by 1.6 mmol/L for every 100 mg/dL (5.6 mmol/L) increase in serum glucose, although this may somewhat underestimate the impact of glucose to decrease serum $[Na^+]$.

Pseudohyponatremia occurs when the solid phase of serum (usually 6%–8%) is increased by large increments in either lipids or proteins (e.g., in hypertriglyceridemia and paraproteinemias). Serum osmolality is normal in pseudohyponatremia. This false result occurs because the usual method, indirect ion-selective potentiometry, measures the concentration of sodium in whole serum and not just the liquid phase, in which the concentration of sodium is 150 mmol/L. Many laboratories use direct ion-selective potentiometry, giving the true aqueous sodium-related osmolality, so if there is a divergence in the plasma sodium between the indirect and direct methods of measurement, a diagnosis of pseudohyponatremia becomes probable. In the absence of a direct-reading potentiometer, an estimate of serum water may be obtained from the following formula[10]:

$$\text{Serum water content (\%)} = 99.1 - (0.1 \times L) - (0.07 \times P)$$

where *L* and *P* refer to the total lipid and protein concentration (in g/L), respectively. For example, if the formula reveals that serum water is 90% of the serum sample rather than the normal 93% (which yields a serum sodium concentration of 140 mmol/L as 150 × 0.93 = 140), the concentration of measured sodium would be expected to decrease to 135 mmol/L (150 × 0.90).

ESTIMATION OF TOTAL BODY WATER

In normal individuals, total body water is approximately 60% of body weight (50% in females and obese individuals). With hyponatremia or hypernatremia, the change in total body water can be calculated from the serum $[Na^+]$ by the following formula:

$$\text{Water excess} = 0.6W \times \left(1 - \frac{[Na^+]_{obs}}{140}\right)$$

$$\text{Water deficit} = 0.6W \times \left(\frac{[Na^+]_{obs}}{140} - 1\right)$$

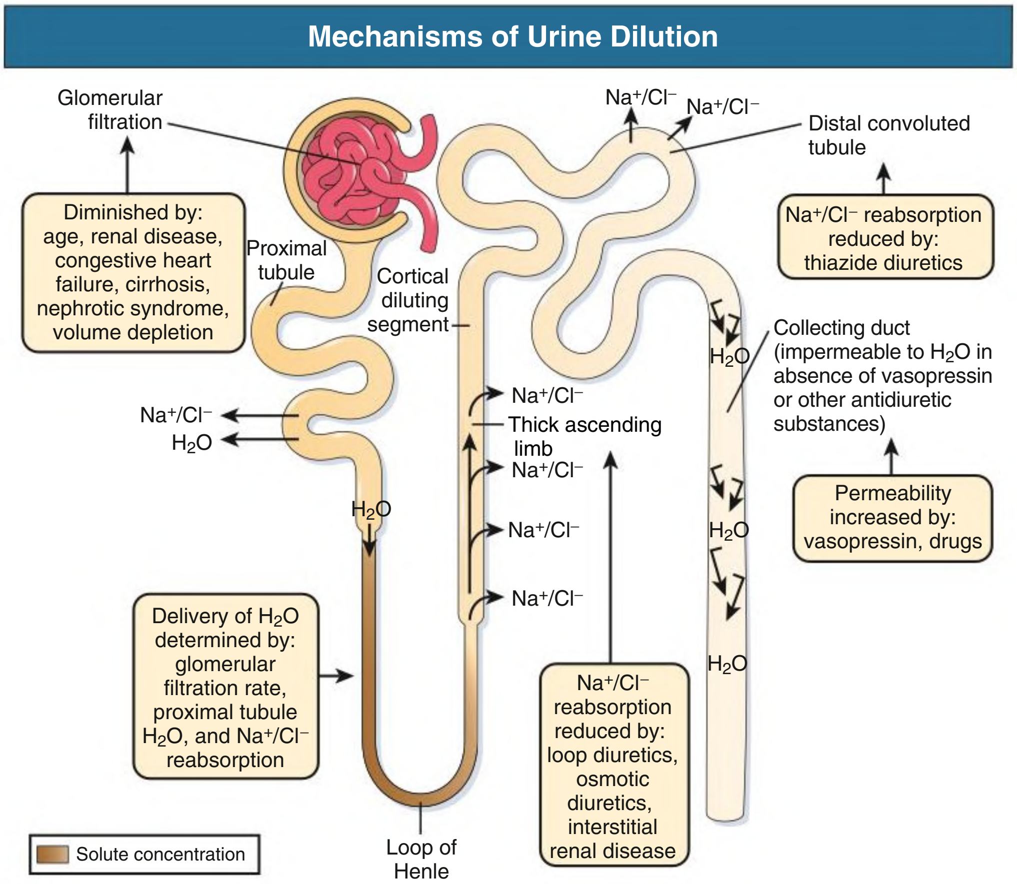

Fig. 9.4 Mechanisms of Urine Dilution. Normal determinants of urinary dilution and disorders causing hyponatremia. (Modified from Cogan MC. Normal water homeostasis. In: Cogan MC, ed. *Fluid and Electrolytes.* Lange; 1991:98–106.)

where $[Na^+]_{obs}$ is observed sodium concentration (in mmol/L) and *W* is body weight (in kilograms). By use of this formula, a change of 10 mmol/L in the serum $[Na^+]$ in a 70-kg individual is equivalent to a change of 3 L in free water.

HYPONATREMIC DISORDERS

Hyponatremia is defined as serum $[Na^+]$ of less than 135 mmol/L. If translocational and pseudohyponatremia are not responsible for the hyponatremia, the decrease in serum sodium concentration reflects a low serum osmolarity, also designated as hypotonicity. The underlying cause of hypotonic hyponatremia is a disturbance in the urinary diluting mechanism.[11] Fig. 9.4 depicts the normal diluting process and the sites at which it can be disrupted so as to impair water excretion, causing its retention and culminating in hyponatremia. A diminished glomerular filtration rate (GFR) and/or an increase in proximal tubular fluid reabsorption decrease distal delivery of filtrate to the diluting segments of the nephron, quantitatively limiting maximal water excretion. Also, hyponatremia may result from a decrement in Na^+-Cl^- transport from the thick ascending limb of the loop of Henle [TAL] or distal convoluted tubule, critical processes in the generation of a dilute tubular fluid. Most frequently, hyponatremia results from nonosmotic vasopressin secretion despite the presence of serum hypo-osmolality. This renders the collecting duct water permeable, thereby not allowing for the excretion of a maximally dilute urine.

Etiology and Classification of Hyponatremia

Once the patient is ascertained as being hypo-osmolar, the next step is to determine whether the individual is hypovolemic, euvolemic, or hypervolemic (Fig. 9.5).

Hypovolemia: Hyponatremia Associated With Decreased Total Body Sodium

A patient with hypovolemic hyponatremia has both a total body Na^+ and a water deficit, with the Na^+ deficit exceeding the water deficit. This occurs in patients with high gastrointestinal (GI) and renal losses of water and solute accompanied by free water or hypotonic fluid intake. The underlying mechanism is the nonosmotic release of vasopressin stimulated by volume contraction, which maintains vasopressin secretion despite the hypotonic state. Measurement of urine $[Na^+]$ is a useful tool in helping to diagnose these conditions (see Fig. 9.5).

Gastrointestinal and third-space sequestered losses. The kidney responds to volume contraction by conserving Na^+ and Cl^-. A similar response is observed in burn patients and patients with sequestration of fluids in third spaces, as in the peritoneal cavity with peritonitis or pancreatitis or in the bowel lumen with ileus. In all these, urine $[Na^+]$ is usually less than 10 mmol/L, and the urine is hyperosmolar. An exception occurs in patients with vomiting and metabolic alkalosis. Here, increased bicarbonate ion ($HCO3^-$) excretion obligates simultaneous cation excretion, resulting in a urine $[Na^+]$ that may exceed 20 mmol/L despite severe volume depletion; however, the urine $[Cl^-]$ remains less than 10 mmol/L. Sodium

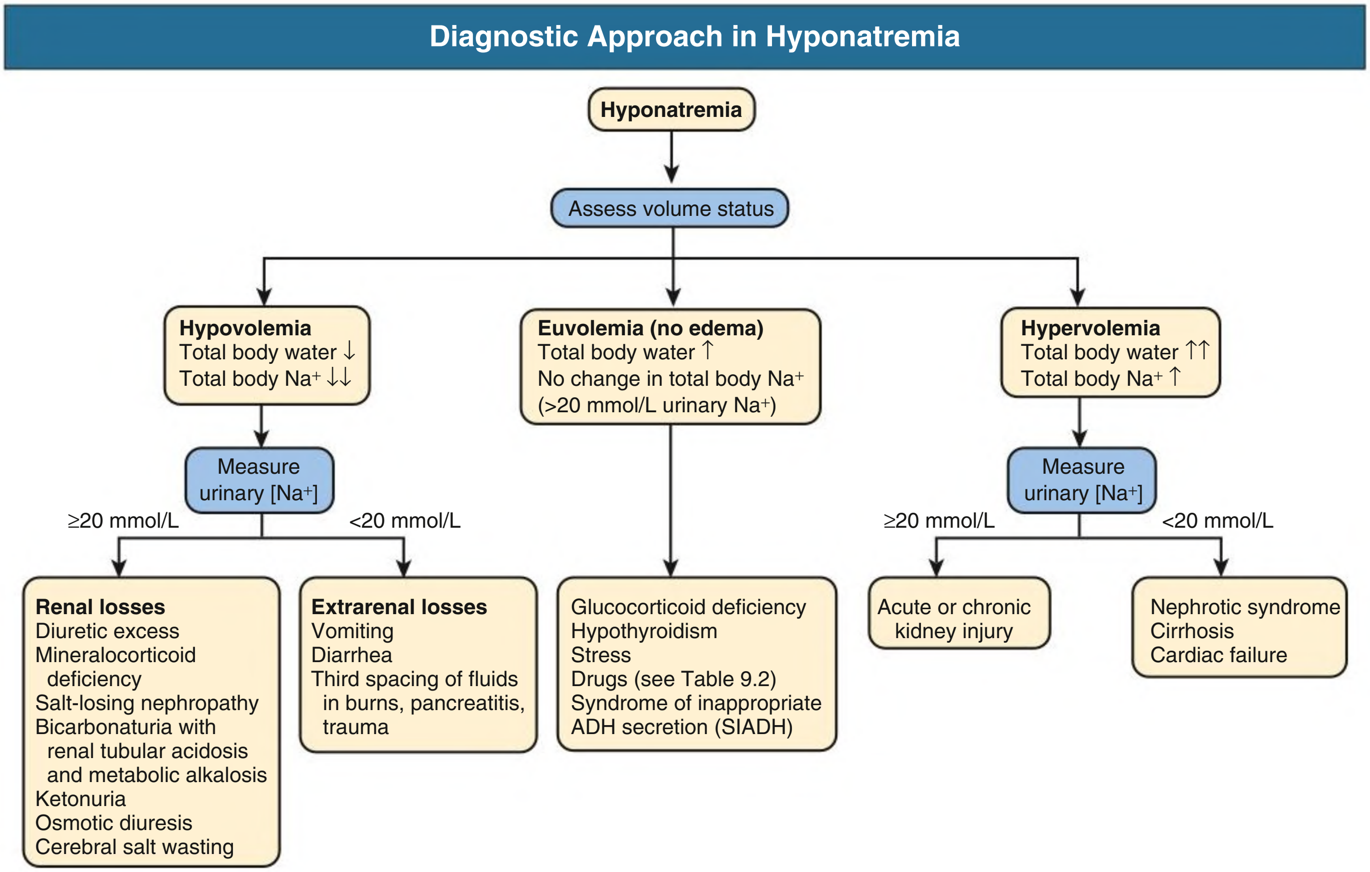

Fig. 9.5 Algorithm for diagnostic assessment of the patient with hyponatremia. (Modified from Halterman R, Berl T. Therapy of dysnatremic disorders. In: Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension.* Saunders; 1999:257–269.)

conservation is also impaired in CKD that may also result in high urine $[Na^+]$ despite volume depletion.

Diuretics. Diuretic use is one of the most common causes of hypovolemic hyponatremia associated with a high urine $[Na^+]$. Loop diuretics inhibit Na^+-Cl^- reabsorption in the thick ascending limb (TAL), impairing the generation of a hypertonic medullary interstitium. Therefore, despite increased AVP secretion, responsiveness to AVP is diminished and free water is excreted. In contrast, thiazide diuretics act in the distal tubule where they interfere with urine dilution, limiting free water excretion. Hyponatremia usually occurs within 14 days of initiation of therapy, with manifestation in approximately one-third of patients within 5 days. Underweight women and elderly patients are most susceptible. Postulated mechanisms for diuretic-induced hyponatremia include the following:

- Hypovolemia-stimulated AVP release and decreased fluid delivery to the diluting segment
- Impaired water excretion through interference with maximal urinary dilution in the cortical diluting segment
- K^+ depletion, directly stimulating water intake by alterations in osmoreceptor sensitivity and increasing thirst

Water retention can mask the physical findings of hypovolemia, making the patients with diuretic-induced hyponatremia appear euvolemic.

Salt-losing nephropathy. A salt-losing state may occur in patients with advanced CKD (GFR < 15 mL/min), particularly from interstitial disease. It is characterized by hyponatremia and hypovolemia. In proximal (type 2) renal tubular acidosis, there is renal Na^+ and K^+ wasting despite only moderately reduced GFR and bicarbonaturia further obligating urine Na^+ excretion.

Mineralocorticoid deficiency. Mineralocorticoid deficiency is characterized by hyponatremia with ECF volume contraction, urine $[Na^+]$ greater than 20 mmol/L, and high serum K^+, urea, and creatinine. Decreased ECF volume provides the nonosmotic stimulus for AVP release.

Osmotic diuresis. An osmotically active, nonreabsorbable solute obligates the renal excretion of Na^+ and results in volume depletion. In the face of continuing water intake, the diabetic patient with severe glycosuria, the patient with a urea diuresis after relief of urinary tract obstruction, and the patient with mannitol diuresis all undergo urinary losses of Na^+ and water, leading to hypovolemia and hyponatremia. Urine $[Na^+]$ is typically greater than 20 mmol/L. The ketone bodies β-hydroxybutyrate and acetoacetate also obligate urinary electrolyte losses and aggravate the renal Na^+ wasting seen in diabetic ketoacidosis, starvation, and alcoholic ketoacidosis.

Cerebral salt wasting. Cerebral salt wasting is a syndrome described primarily in patients with subarachnoid hemorrhage. The primary defect is salt wasting from the kidneys with subsequent volume contraction, which stimulates vasopressin release. The exact mechanism is not understood, but it is postulated that brain natriuretic peptide increases urine volume and Na^+ excretion. Serum uric acid is often low, similar to that observed in syndrome of inappropriate antidiuretic hormone (SIADH; see later). The diagnosis requires evidence of inappropriate sodium losses and reduced effective blood volume, usually with signs of orthostatic hypotension. These criteria are rarely fulfilled, suggesting that cerebral salt wasting is overdiagnosed.

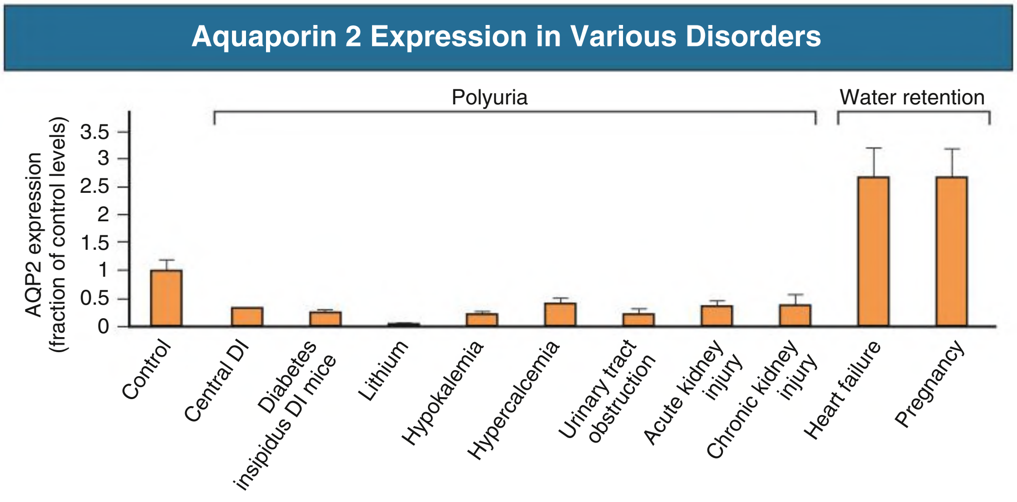

Fig. 9.6 Changes In Aquaporin 2 (Aqp2) Expression Seen in Association With Different Water Balance Disorders. Levels are expressed as a fraction (percentage) of control levels. AQP2 expression is reduced, sometimes dramatically, in a wide range of hereditary and acquired forms of diabetes insipidus (DI) characterized by different degrees of polyuria. Conversely, congestive heart failure and pregnancy are conditions associated with increased expression of AQP2 levels and excessive water retention. (Modified from Nielsen S, Knepper MA, Kwon TH. Regulation of water balance. Urine concentration and dilution. In: Schrier RW, ed. *Diseases of the Kidney and Urinary Tract.* 8th ed. Lippincott Williams & Wilkins; 2007:96–123.)

Hypervolemia: Hyponatremia Associated With Increased Total Body Sodium

Hyponatremia can also develop in hypervolemic states, such as congestive heart failure (CHF), nephrotic syndrome, and cirrhosis, in which the total body water is increased more than total body Na^+ (see Fig. 9.5 and Chapter 8).

Congestive heart failure. Edematous patients with CHF have reduced effective intravascular volume as a result of decreased systemic mean arterial pressure (MAP) and cardiac output. This reduction is sensed by aortic and carotid baroreceptors activating nonosmotic pathways, resulting in vasopressin release. In addition, the relative "hypovolemic" state stimulates the renin-angiotensin axis and increases norepinephrine production, which, in turn, decreases GFR. This causes an increase in proximal tubular reabsorption and a decrease in water delivery to the distal tubule.

The neurohumorally mediated decrease in delivery of tubular fluid to the distal nephron and an increase in vasopressin secretion mediate hyponatremia by limiting Na^+-Cl^- and water excretion. In addition, low cardiac output and high Ang II levels are potent stimuli of thirst. There is also excessive intracellular targeting of AQP2 to the apical cell membrane of the collecting duct, most likely from high AVP levels (Fig. 9.6).[12]

As cardiac function improves with afterload reduction, plasma vasopressin decreases, with concomitant improvement in water excretion. The degree of hyponatremia has also been correlated with the severity of cardiac disease and with patient survival; serum [Na^+] less than 125 mmol/L reflects severe CHF.

Hepatic failure. Patients with cirrhosis and hepatic insufficiency also have increased ECF volume (ascites, edema). Because of splanchnic venous dilation, they have increased plasma volume. Cirrhotic patients have an increased cardiac output because of multiple arteriovenous fistulas in their alimentary tract, lungs, and skin. Vasodilation and arteriovenous fistulas cause a decrease in MAP. As the severity of cirrhosis increases, there are progressive increases in plasma renin, norepinephrine, AVP, and endothelin, and an associated decline in MAP and serum [Na^+]. In experimental models, expression of AQP2 is upregulated in collecting ducts.[7]

Nephrotic syndrome. In some patients with nephrotic syndrome, especially those with minimal change disease, low plasma oncotic pressure from hypoalbuminemia alters Starling forces, leading to intravascular volume contraction and stimulation of AVP with hyponatremia. In contrast, most nephrotic patients have a renal defect in sodium excretion resulting in increased effective circulating volume. Hyponatremia may still occur in these latter conditions. In experimental models of nephrotic syndrome, expression of AQP2, AQP3, and UT-A1 are downregulated in the collecting duct (which would have countering effects), but so are the sodium transporters, NKCC2, NHE3, and Na-K-ATPase, in the TAL, resulting in hyponatremia.[7]

Advanced chronic kidney disease. Patients with severely reduced GFR, either acute or chronic, have a profound increase in fractional excretion of Na^+ to maintain normal salt balance given the overall decreased number of functioning nephrons. Edema usually develops when the Na^+ ingested exceeds the capacity of the kidneys to excrete this load. Likewise, if water intake exceeds threshold, there is positive water balance and hyponatremia. At a GFR of 5 mL/min, only 7.2 L of filtrate is formed daily. Approximately 30%, or 2.2 L, of this filtered fluid will reach the diluting segment of the nephron, which is therefore the maximum solute-free water that can be excreted daily. As GFR progressively declines, a defect in renal concentration precedes a disorder in urinary dilution. The excretion of free water as a function of remaining glomeruli is well maintained until kidney failure is very advanced.[13]

Older age. Aging results in 20% reduction in maximum urine osmolality in people older than 60 years. The concentrating defect is not related to a decrease in GFR or an abnormality in AVP secretion.

Euvolemia: Hyponatremia Associated With Normal Total Body Sodium

Euvolemic hyponatremia is the most common dysnatremia in hospitalized patients. These patients have no physical signs of increased or decreased total body Na^+.

Glucocorticoid deficiency. Cortisol exhibits a negative feedback on ACTH and vasopressin production so primary or secondary glucocorticoid deficiency leads to impaired water excretion from excess circulating vasopressin. This can be corrected by physiologic doses of corticosteroids but not by volume expansion. Hyponatremia may be enhanced by reduced renal blood flow and decreased distal fluid delivery to the diluting segments of the nephron.

Hypothyroidism. Hyponatremia occurs in patients with severe hypothyroidism, who usually meet the clinical criteria for myxedema coma. A decrease in cardiac output leads to nonosmotic release of AVP. A reduction in GFR leads to diminished free water excretion through decreased distal delivery to the distal nephron. The exact mechanisms are unclear. In patients with untreated hypothyroidism who have moderately severe disease, a vasopressin-independent mechanism is suggested by normal suppression of vasopressin after water loading. However, in patients with more advanced hypothyroidism, elevated vasopressin levels are reported in the basal state and after a water load. Hyponatremia is readily reversed by hormonal replacement treatment.

Psychosis. Patients with acute psychosis may develop hyponatremia. Psychogenic drugs, particularly selective serotonin reuptake inhibitors (SSRIs), are associated with hyponatremia, but psychosis can cause hyponatremia independently.[14] The pathophysiologic process involves an increased thirst perception, a mild defect in osmoregulation that causes vasopressin to be secreted at lower osmolality, and an enhanced renal response to vasopressin. Individuals with self-induced water intoxication may also be more prone to rhabdomyolysis.

Postoperative hyponatremia. Postoperative hyponatremia mainly is a result of excessive infusion of electrolyte-free water (hypotonic saline or 5% dextrose in water) and the presence of AVP, which prevents water excretion. Hyponatremia can also occur despite infusion with near-isotonic (normal) saline within 24 hours of induction of anesthesia.[15] Young premenstrual girls are at higher risk of acute postoperative hyponatremia accompanied by cerebral edema. The mechanism has not been fully elucidated, and the patients at highest risk cannot be prospectively identified. Nevertheless, hypotonic fluids should be avoided after surgery, isotonic fluids minimized, and serum [Na^+] checked if hyponatremia is suspected (e.g., symptoms of headache or nausea).

Exercise-induced hyponatremia. Hyponatremia is seen in long-distance runners. A study at a marathon race associated increased risk of hyponatremia with body mass index less than 20 kg/m^2, running time exceeding 4 hours, and greatest weight gain.[16] A study in ultramarathon runners showed elevated AVP despite normal or low serum [Na^+]. The mechanism may be because of an exuberant AVP response in relation to exercise-induced dehydration.

Drugs causing hyponatremia. Drug-induced hyponatremia is becoming the most common cause of hyponatremia.[14] Thiazide diuretics and SSRIs are the most commonly implicated medications. Hyponatremia can be mediated by AVP analogs, such as desmopressin (DDAVP, 1-desamino-D-arginine AVP), that enhance AVP release and agents potentiating the action of AVP. In other cases, the mechanism is unknown (Table 9.2). Desmopressin for nocturia in elderly patients and enuresis in young persons has caused hyponatremia in some of these subjects, so it is not routinely used to treat these conditions. Desmopressin for nocturia should be used only when the number of voidings are not tolerable and are debilitating (>2/night). Upon awakening, the drug is likely to be still active for several more hours, requiring great attention to water restriction. Hyponatremia may also result from use of intravenous immunoglobulin (IVIG).[17] The mechanism of IVIG-associated hyponatremia is multifactorial, involving pseudohyponatremia (because of increases in serum protein concentration), translocation (because of sucrose in the solution), and true dilutional hyponatremia (because of the retention of water, particularly when associated with acute kidney injury [AKI]).[17]

TABLE 9.2 Drugs Associated With Hyponatremia[a]

Vasopressin Analogs	Drugs That Potentiate Renal Action of Vasopressin
Desmopressin (DDAVP)	Chlorpropamide
Oxytocin	Cyclophosphamide
	Nonsteroidal anti-inflammatory drugs (NSAIDs)
	Acetaminophen
Drugs That Enhance Vasopressin Release	**Drugs That Cause Hyponatremia by Unknown Mechanisms**
Chlorpropamide	Haloperidol
Clofibrate	Fluphenazine
Carbamazepine-oxcarbazepine	Amitriptyline
Vincristine	Thioridazine
Nicotine	Fluoxetine
Narcotics	Methamphetamine (MDMA, “ecstasy”)
Antipsychotics/antidepressants (SSRIs)	Intravenous immunoglobulin (IVIG)
Ifosfamide	

Terms in italics are the most common causes.
[a]Not including diuretics.
MDMA, 3,4-Methylenedioxymethamphetamine; *SSRIs,* selective serotonin reuptake inhibitors.
From Liamis G, Elisaf M. Hyponatremia induced by drugs. In: Simon E, ed. *Hyponatremia:* Springer; 2013:111–126.

Syndrome of inappropriate antidiuretic hormone secretion. Despite being the most common cause of hyponatremia in hospitalized patients, SIADH is a diagnosis of exclusion. A defect in osmoregulation causes AVP to be inappropriately stimulated, leading to urine concentration (Table 9.3). A few causes deserve special mention. Central nervous system (CNS) disturbances such as hemorrhage, tumors, infections, and trauma cause SIADH by excess AVP secretion. Small cell lung cancers, cancer of the duodenum and pancreas, and olfactory neuroblastoma cause ectopic production of AVP. Idiopathic cases of SIADH are unusual except in elderly patients, in whom hyponatremia is frequently multifactorial,[18] but in whom as many as 10% have abnormal AVP secretion without known cause.

Several patterns of abnormal AVP release have emerged from studies of patients with clinical SIADH[19] and have now been reproduced employing the simpler, more reliable, and more stable copeptin assay.[20] In one-third of patients with SIADH, AVP release varies appropriately with serum [Na^+] but begins at a lower threshold of serum osmolality, implying a “resetting of the osmostat.” Ingestion of free water then leads to water retention to maintain the serum [Na^+] at a new lower level, usually 125 to 130 mmol/L. In two-thirds of patients, AVP release does not correlate with serum [Na^+], but a solute-free urine cannot be excreted. Therefore, ingested water is retained, giving rise to moderate nonedematous volume expansion and dilutional hyponatremia. In about 10% of patients, AVP levels are not measurable, suggesting that the *syndrome of inappropriate antidiuresis* (SIAD) is a more accurate term.[19] It was suggested that such patients may have a nephrogenic syndrome of antidiuresis, and a gain-of-function mutation in the AVP receptor. However, none of the six patients who had unmeasurable copeptin (a surrogate marker for AVP) levels had an identifiable gain of function mutation.[20]

The diagnostic criteria for SIADH are summarized in Box 9.1.[1] Plasma AVP may be in the “normal” range (up to 10 ng/L), but this

TABLE 9.3 Causes of Syndrome of Inappropriate Antidiuretic Hormone Secretion

Carcinomas	Pulmonary Disorders	Nervous System Disorders	Other
Bronchogenic carcinoma	*Viral pneumonia*	*Encephalitis (viral, bacterial)*	*Human immunodeficiency virus (HIV) infection; acquired immunodeficiency syndrome (AIDS)*
Carcinoma of duodenum	*Bacterial pneumonia*	*Meningitis (viral, bacterial, tuberculous, fungal)*	*Idiopathic (elderly)*
Carcinoma of pancreas	*Pulmonary abscess*	*Head trauma*	Prolonged exercise
Thymoma	*Tuberculosis*	*Brain abscess*	
Carcinoma of stomach	Aspergillosis	*Brain tumors*	
Lymphoma	Positive-pressure ventilation	Guillain-Barré syndrome	
Ewing sarcoma	Asthma	Acute intermittent porphyria	
Carcinoma of bladder	Pneumothorax	Subarachnoid hemorrhage or subdural hematoma	
Carcinoma of prostate	Mesothelioma	Cerebellar and cerebral atrophy	
Oropharyngeal tumor	Cystic fibrosis	Cavernous sinus thrombosis	
Carcinoma of ureter		Neonatal hypoxia	
		Hydrocephalus	
		Shy-Drager syndrome	
		Rocky Mountain spotted fever	
		Delirium tremens	
		Cerebrovascular accident (stroke; cerebral thrombosis or hemorrhage)	
		Acute psychosis	
		Peripheral neuropathy	
		Multiple sclerosis	

Terms in italics are the most common causes.
From Thurman JM, Berl T. Therapy of dysnatremic disorders. In: Wilcox CS, ed. *Therapy in Nephrology and Hypertension*. 3rd ed. Saunders; 2008:337–352.

BOX 9.1 Diagnostic Criteria for Syndrome of Inappropriate Antidiuretic Hormone Secretion

Essential Diagnostic Criteria
- Decreased extracellular fluid effective osmolality (270 mOsm/kg H_2O)
- Inappropriate urine concentration (>100 mOsm/kg H_2O)
- Clinical euvolemia
- Elevated urine Na^+ concentration under conditions of normal salt and water intake
- Absence of adrenal, thyroid, pituitary, or renal insufficiency or diuretic use

Supplemental Criteria
- Abnormal water-load test result (inability to excrete at least 90% of a 20-mL/kg water load in 4 hours and/or failure to dilute urine osmolality to <100 mOsm/kg)
- Serum vasopressin level inappropriately elevated relative to the serum osmolality
- No significant correction of serum Na^+ level with volume expansion, but improvement after fluid restriction
- Hypouricemia and elevated fractional excretion of uric acid

Modified from Verbalis J. The syndrome of inappropriate antidiuretic hormone secretion and other hypo-osmolar disorders. In: Schrier RW, ed. *Diseases of the Kidney and Urinary Tract.* 9 ed. Philadelphia: Lippincott Williams & Wilkins; 2013:2012-54.

is inappropriate given the hypo-osmolar state. The measurement of plasma AVP is rarely needed because the urinary osmolality provides an excellent surrogate bioassay. Thus, a hypertonic urine (>300 mOsm/kg H_2O) provides strong evidence for the presence of AVP in the circulation because such urinary tonicities are unattainable in its absence. Likewise, a urinary osmolality lower than 100 mOsm/kg H_2O reflects the virtual absence of the hormone. Urinary osmolarities in the range of 100 to 300 mOsm/kg H_2O can occur in the presence or absence of AVP. A decrease in serum uric acid concentration associated with a high fractional excretion (>10%) is frequently encountered in the patient with SIADH.

Clinical Manifestations of Hyponatremia

Most patients with serum [Na^+] greater than 125 mmol/L are asymptomatic. In patients with less than 125 mmol/L, but more specifically less than 120 mmol/L, which is typically the definition of "severe hyponatremia," headache, yawning, lethargy, nausea, reversible ataxia, psychosis, seizures, and coma may occur as a result of cerebral edema. Neurologic symptoms in a hyponatremic patient call for immediate attention and treatment.

"Moderate hyponatremia" and "mild hyponatremia" are normally defined with ranges from 120 to 129 mmol/L and 130 to 134 mmol/L, respectively. In the setting of nonsevere disease, complications are less frequently manifested.

The catastrophic consequence of significant hyponatremia is cerebral edema so severe that there is increased intracerebral pressure, tentorial herniation, respiratory depression, and death. These events occur with rapid development of hyponatremia, usually in hospitalized postoperative patients receiving diuretics or hypotonic fluids. Untreated severe hyponatremia has a mortality rate as high as 50%, and it is in this group where the complications from not only the hyponatremia itself but also its overcorrection are most commonly seen.

Cerebral Edema

The development of cerebral edema largely depends on the cerebral adaptation to hypotonicity. Decreases in extracellular osmolality cause movement of water into cells, increasing intracellular volume and causing tissue edema. The water channel AQP4 appears to play a key role in the movement of water across the blood-brain barrier. AQP4 knockout mice are protected from hyponatremic brain

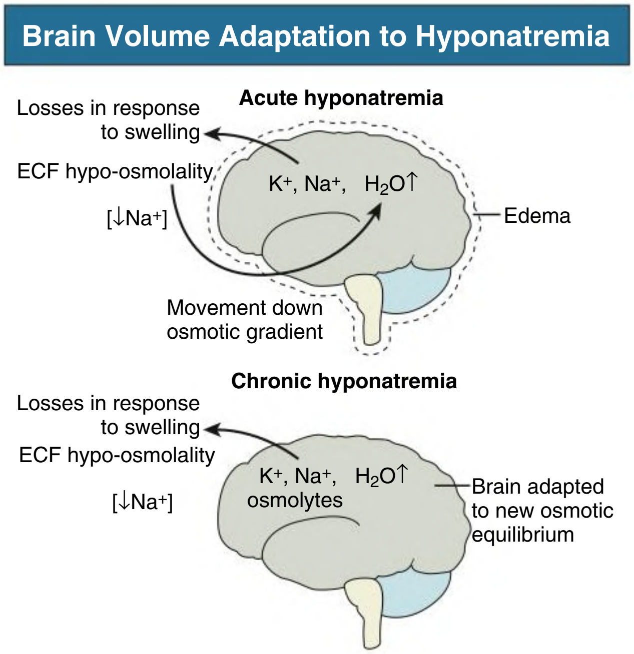

Fig. 9.7 Brain Volume Adaptation to Hyponatremia. During acute hyponatremia, water enters the brain to establish osmotic equilibrium with the ECF. As an acute adaptive change, NaCl exits from the brain interstitial space, followed by loss of potassium from cells several hours later. In chronic hyponatremia, the brain loses osmolytes, which lead to further water losses from the brain and an almost full restoration of brain water to levels marginally greater than baseline. (Modified from Verbalis J. The syndrome of inappropriate antidiuretic hormone secretion and other hypo-osmolar disorders. In: Schrier RW, ed. *Diseases of the Kidney and Urinary Tract*. 9th ed. Lippincott Williams & Wilkins; 2013:2012–2054.)

swelling, whereas animals overexpressing AQP4 have exaggerated brain swelling.[21] Cellular edema within the fixed confines of the cranium increases intracranial pressure, leading to the neurologic syndrome. In most hyponatremic patients, mechanisms of volume regulation prevent cerebral edema.

Within 1 to 3 hours after hyponatremia develops, a decrease in cerebral extracellular volume occurs by movement of fluid into the cerebrospinal fluid, which is then shunted back into the systemic circulation. Loss of extracellular solutes Na^+ and Cl^- can occur as early as 30 minutes after the onset of hyponatremia (Fig. 9.7). If hyponatremia persists for longer than 3 hours, the brain adapts by losing cellular osmolytes, including K^+ and organic solutes, which tends to lower the osmolality of the brain, resulting in water losses. Thereafter, if hyponatremia persists, other organic osmolytes, such as phosphocreatine, myoinositol, and amino acids (e.g., glutamine, taurine), are lost, greatly decreasing cerebral swelling. As a result of these adaptations, most patients have minimal symptoms despite severe hyponatremia ($[Na^+] < 120$ mmol/L).

Certain patients are at increased risk for development of acute cerebral edema in the course of hyponatremia[12] (Table 9.4). Hospitalized premenstrual girls with hyponatremia are more symptomatic and more likely to have complications of therapy than postmenopausal women or men. This increased risk of cerebral edema is independent of the rate of development or the magnitude of hyponatremia. The best management of these patients is to avoid the administration of hypotonic fluids in the postoperative setting. Children are particularly vulnerable to the development of acute cerebral edema, perhaps because of a relatively high ratio of brain to skull volume.

TABLE 9.4 Persons at Risk for Neurologic Complications With Hyponatremia

Acute Cerebral Edema	Osmotic Demyelination Syndrome (Central Pontine Myelinolysis)
Postoperative menstruating women	Liver transplant recipients
Elderly women taking thiazides	Alcoholic patients
Children	Malnourished patients
Patients with polydipsia secondary to psychiatric disorders	Hypokalemic patients
	Burn patients
Hypoxemic patients	Elderly women taking thiazides
Marathon runners	Hypoxemic patients
	Severe hyponatremia ($[Na^+] < 105$ mmol/L)

Patient groups at risk for acute cerebral edema and central pontine myelinolysis (osmotic demyelination).
From Thurman JM, Berl T. Therapy of dysnatremic disorders. In: Wilcox CS, ed. *Therapy in Nephrology and Hypertension*. 3rd ed. Elsevier; 2008:337–352.

Osmotic Demyelination

Another neurologic syndrome can occur in hyponatremic patients as a complication of correction of hyponatremia. Osmotic demyelination most often affects the central pons and is therefore also termed *central pontine myelinolysis* (CPM). It occurs at all ages; Table 9.4 lists patients at highest risk. Osmotic demyelination syndrome is especially common after liver transplantation, with a reported incidence of 13% to 29% at autopsy. The risk of CPM is related to the severity and chronicity of the hyponatremia. It rarely occurs with serum $[Na^+]$ greater than 120 mmol/L or a short duration of hyponatremia (<48 hours). The symptoms are biphasic. Initially, there is a generalized encephalopathy associated with rapid correction of serum $[Na^+]$. At 2 to 3 days after correction, the patient displays behavioral changes, cranial nerve palsies, and progressive weakness, culminating in quadriplegia and a “locked-in” syndrome. T2-weighted magnetic resonance imaging shows nonenhancing and hyper-intense pontine and extrapontine lesions. These lesions may not appear until 2 weeks after development, so a diagnosis of myelinolysis should not be excluded if the imaging is initially normal.

The pathogenesis of osmotic demyelination syndrome is uncertain; one suggestion is that sodium-coupled amino acid transporters (e.g., SNAT2) are downregulated by hypotonicity, thereby delaying the return of osmolytes to the brain, rendering it more sensitive to the correction of hyponatremia.[22] Although $[Na^+]$ and $[K^+]$ return to normal in a few hours, osmotically active solutes require several days to return to normal levels. This temporary imbalance causes cerebral dehydration and can lead to a potential breakdown of the blood-brain barrier. Astrocytes appear to be an early target of the disease, activating microglial cells resulting in the expression of proinflammatory cytokines.[23]

Although CPM was originally considered to be uniformly fatal, a substantial number of patients can have some neurologic recovery, even with severe symptoms at onset, suggesting there are reversible forms of osmotic demyelination.

Treatment of Hyponatremia

Symptoms and duration of hyponatremia determine treatment. Acutely hyponatremic patients (developing within 48 hours) are at great risk for permanent neurologic sequelae from cerebral edema if the hyponatremia remains uncorrected. Patients with chronic

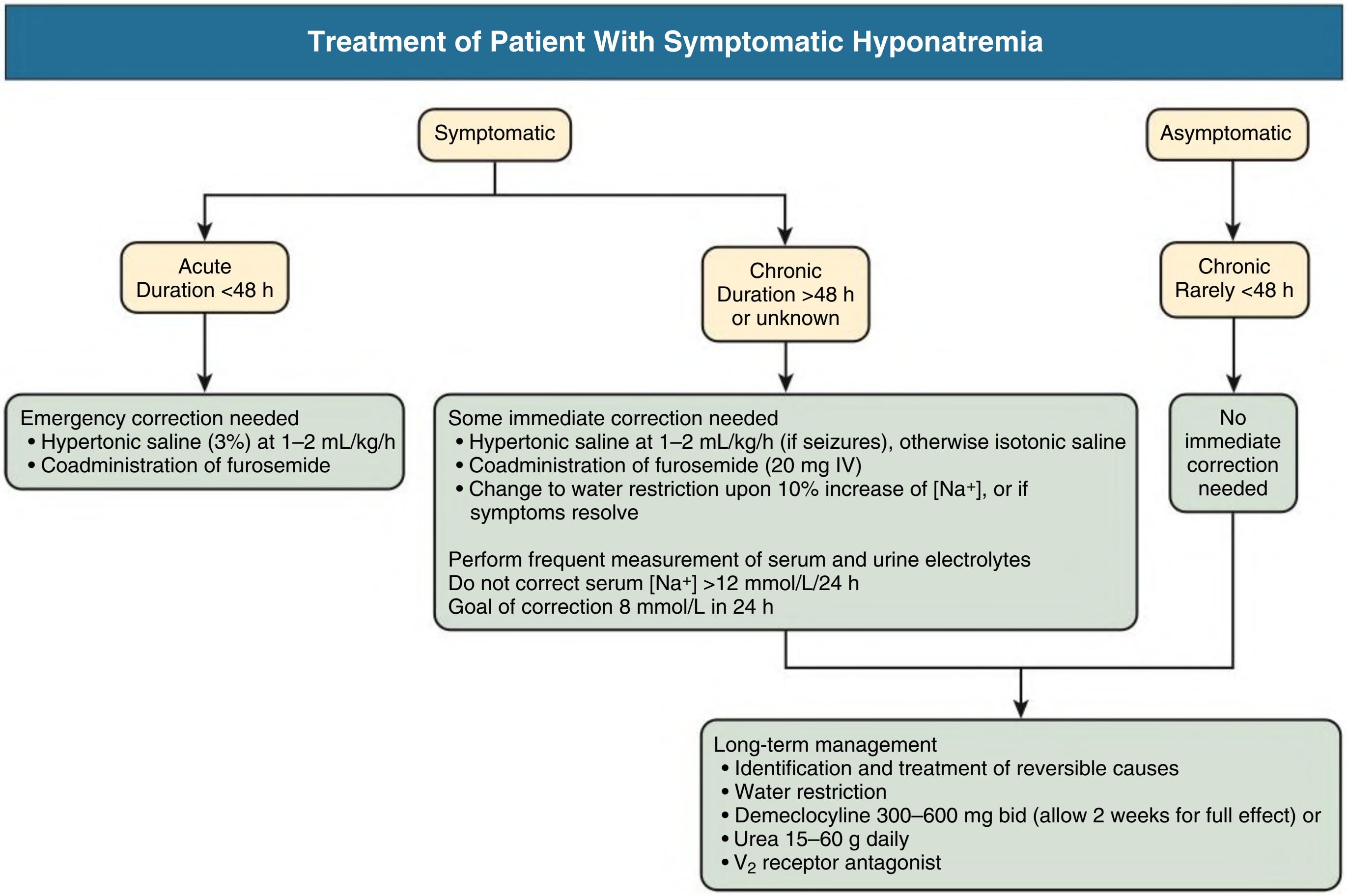

Fig. 9.8 Algorithm for management of the patient with symptomatic (and "asymptomatic") hyponatremia. *IV*, Intravenous. (Modified from Thurman JM, Berl T. Therapy of dysnatremic disorders. In: Wilcox CS, ed. *Therapy in Nephrology and Hypertension.* 3rd ed. Elsevier; 2008:337–352.)

hyponatremia are at risk for osmotic demyelination if the hyponatremia is corrected too rapidly.

Acute Symptomatic Hyponatremia

Acute symptomatic hyponatremia with seizures or other neurologic manifestations almost always develops in hospitalized patients receiving hypotonic fluids (Fig. 9.8). Treatment should be prompt because the risk of acute cerebral edema far exceeds the risk of osmotic demyelination. The cell volume adaptive response, whereby the brain decreases its water content in acute hyponatremia, may be inhibited by female hormones, possibly explaining the large female predominance. A contribution of hypoxia also may be important, because when hypoxia is combined with hyponatremia in experimental animals, the volume adaptive response is abrogated, resulting in brain edema and increased mortality.[24] Because the neurologic complications associated with acute symptomatic hyponatremia are devastating, patients require prompt treatment with 3% NaCl.[12] A rapid increment of 4 to 6 mmol/L within the first 6 hours appears to be sufficient to reverse cerebral edema; correction to normal levels is unnecessary.[25] Initial treatment should entail an infusion of 3% NaCl 1 to 2 mL/kg over 60 minutes or a 100 mL bolus over 10 minutes followed by two further doses to raise the plasma sodium to the specified level. Administration of a loop diuretic enhances free water excretion and hastens the normalization of serum [Na^+]. If the patient presents with severe neurologic symptoms, such as seizures, obtundation, or coma, 3% NaCl may be infused at higher rates (4–6 mL/kg/hr). Patients should be monitored carefully for changes in neurologic and pulmonary status, and serum electrolytes should be checked every 2 hours.

Chronic Symptomatic Hyponatremia

If the hyponatremia has taken more than 48 hours to evolve or if the duration is not known, correction should be cautious (see Fig. 9.8). Controversy exists as to whether it is the rate of correction or the magnitude of correction of hyponatremia that predisposes to neurologic complications. It is difficult to dissociate these two variables because a rapid correction rate is usually accompanied by a greater absolute magnitude of correction during a given time. Three important principles guide treatment:

1. Because cerebral water is increased by approximately 10% in severe chronic hyponatremia, the goal is to increase the serum Na^+ level by 10%, or about 10 mmol/L, in the first 24 hours.
2. A correction rate of 1.0 to 1.5 mmol/L in any given hour should not be exceeded.
3. The goal of treatment is an increase of 6 to 10 mmol/L every 24 hours but do not increase by more than 12 mmol/L every 24 hours and 18 mmol/L per 48 hours.

Numerous formulas have been used to assess the expected changes in serum sodium with various infusions, but the one proposed by Adrogue and Madias is the most widely employed.[26] Although helpful in the initial treatment phase, it does not take into account ongoing renal and extrarenal losses, thus commonly resulting in correction larger than those predicted by the formula. The risk for overcorrection

TABLE 9.5 Treatment of Patients With Chronic Asymptomatic Hyponatremia

Treatment	Mechanism of Action	Dose	Advantages	Limitations
Fluid restriction	Decreases availability of free water	Variable	Effective and inexpensive; not complicated	Noncompliance
Pharmacologic Inhibition of Vasopressin Action				
Lithium	Inhibits kidney's response to vasopressin	900–1200 mg/day	Unrestricted water intake	Polyuria, narrow therapeutic range, neurotoxicity
Demeclocycline	Inhibits kidney's response to vasopressin	300–600 mg twice daily	Effective; unrestricted water intake	Neurotoxicity, polyuria, photosensitivity, nephrotoxicity
V_2 receptor antagonist	Antagonizes vasopressin action	—	Addresses underlying mechanisms	Limited clinical experience
Increased Solute (Salt) Intake				
With furosemide	Increases free water clearance	Titrate to optimal dose; coadminister 2–3 g NaCl	Effective	Ototoxicity, K^+ depletion
With urea	Osmotic diuresis	30–60 g/day	Effective; unrestricted water intake	Polyuria, unpalatable, gastrointestinal symptoms

can be mitigated by the coadministration of DDAVP, thus preventing the excretion of a hypotonic urine.[27] If the aforementioned noted limits are exceeded, relowering of serum sodium can be achieved by the infusion of dextrose and water and the administration of DDAVP.[27]

Chronic "Asymptomatic" Hyponatremia

Patients with chronic hyponatremia are often asymptomatic. However, epidemiologic data consistently show higher mortality rates than matched controls with normal serum sodium levels.[28] Formal neurologic testing frequently reveals subtle impairments, including gait disturbances that reverse with correction of the hyponatremia. This results in an increased risk for falls and fractures.[29] Therefore, even "asymptomatic" patients should be treated in an attempt to restore serum sodium to near-normal levels, particularly if they display gait instability or have sustained a fall. These patients should also be evaluated for hypothyroidism, adrenal insufficiency, and SIADH and should have their medications reviewed.

Fluid restriction. Stopping any medication associated with hyponatremia and fluid restriction is the cornerstone of therapy in patients with chronic asymptomatic hyponatremia (Table 9.5). Intake of all fluids should be restricted, and fructose-containing fluids should be avoided because fructose stimulates vasopressin release independently of osmolarity. This approach is usually successful if patients are compliant. It involves a calculation of the fluid restriction that will maintain a specific serum [Na^+]. The daily osmolar load (OL) and the minimal urinary osmolality $(U_{osm})_{min}$ determine a patient's maximal urine volume (V_{max}), as follows:

$$V_{max} = \frac{OL}{(U_{osm})_{min}}$$

The value of $(U_{osm})_{min}$ is a function of the severity of the diluting disorder. In the absence of circulating vasopressin, it can be as low as 50 mOsm/kg H_2O. In a normal North American diet, the daily osmolar load is approximately 10 mOsm/Kg (700 mOsm for a 70-kg person). Assuming that a patient with SIADH has U_{osm} that cannot be lowered to less than 500 mOsm/kg H_2O, the same osmolar load of 700 mOsm allows only 1.4 L of urine to be excreted daily. Therefore, if the intake exceeds 1.4 L/day, the serum Na^+ concentration will decrease. Measurement of urine [Na^+] and [K^+] can guide the required degree of water restriction.[30] Unless the ratio of urine [Na^+] plus [K^+] over serum [Na^+] is 0.5 or less, the needed water restriction is not likely to be complied with. Likewise, a urinary osmolality greater than 500 mOsm/kg also predicts poor response to water restriction.[1] In a report from a hyponatremia registry, the increment in serum sodium observed in patients placed on water restriction did not achieve statistical significance compared with their baseline admission serum sodium.[31] If the diluting defect is so severe that fluid restriction to less than 1 L is necessary or if the serum Na^+ concentration remains low (<130 mmol/L), an alternative approach to treatment should be considered, such as increasing solute excretion or pharmacologic inhibition of AVP.

Increase solute excretion. If the patient remains unresponsive to fluid restriction, solute intake can be increased to facilitate an obligatory increase in excretion of solute and free water. This can be achieved by increasing oral salt and protein intake in the diet to increase the C_{osm} of the urine. Loop diuretics combined with high sodium intake (2–3 g of additional salt) are effective in the management of hyponatremia. A single dose of a loop diuretic (e.g., 40 mg furosemide) is usually sufficient but should be doubled if the diuresis induced in the first 8 hours is less than 60% of the total daily urine output.

In patients with SIADH, urine osmolality is fixed. In the collecting duct, urea is an effective osmole that increases water loss, thereby decreasing water retention and sodium loss. This permits improvement in hyponatremia and without altering urine concentration. The dose of urea is usually 30 to 60 g/day. The limitations are GI distress and unpalatability.

Pharmacologic inhibition of vasopressin. *Vaptans* are novel oral V_2 receptor antagonists that block vasopressin binding to the collecting duct tubular epithelial cells and increase free water excretion without significantly altering electrolyte excretion.[32–34] Vaptans are effective in the treatment of hyponatremia in euvolemic and hypervolemic patients.[35] Conivaptan, a V_2 and V_{1a} antagonist, is the only vaptan available for IV use,[34] but treatment should be limited to 4 days because it is a potent cytochrome P-450 3A4 (CYP3A4) inhibitor and should generally be avoided in patients at risk for variceal bleeding, given that the antagonism of V_1 receptors would tend to vasodilate existing varices. Tolvaptan, an oral V_2 antagonist, is available at doses of 15 to 60 mg/day. In the tolvaptan trials (TEMPO) in patients with polycystic kidney disease, higher doses were used and some cases of hepatic toxicity, as well as rhabdomyolysis, were encountered. This has led to a Food and Drug Administration (FDA) warning requiring careful monitoring of liver function tests and creatine kinase (CK, CPK) levels. Although the recommendation is to use tolvaptan for less than 30 days, some clinicians are treating chronically for patients with severe SIADH using the lowest dose that can maintain normal serum osmolality so long as liver function tests remain normal. Several newer V_2 selective vaptans (mozavaptan, satavaptan, and lixivaptan) have been made available, but tolvaptan remains the most extensively studied.

An alternative pharmacologic treatment is demeclocycline, 600 to 1200 mg/day given 1 to 2 hours after meals; calcium-, aluminum-, and magnesium-containing antacids should be avoided. Onset of action is usually 3 to 6 days after initiation of treatment. Dose should be titrated to the minimum that keeps serum [Na^+] within the desired range with unrestricted water intake. Demeclocycline can cause photosensitivity or (in children) tooth or bone abnormalities. Polyuria leads to noncompliance, and nephrotoxicity may occur, especially in patients with underlying liver disease. Lithium was previously used to antagonize AVP action but has been superseded by the vaptans and demeclocycline.

Hypovolemic Hyponatremia

When thiazides are prescribed, especially in elderly women, serum [Na^+] should be monitored and water intake restricted. If hyponatremia develops, the thiazide should be discontinued.

Neurologic syndromes directly related to hyponatremia are unusual in hypovolemic hyponatremia because loss of Na^+ and water limits any osmotic shifts in the brain. Restoration of ECF volume with crystalloids or colloids interrupts the nonosmotic release of AVP. AVP antagonists should not be used in these patients.[35]

Hypervolemic Hyponatremia

Congestive heart failure. In patients with CHF, sodium and water restriction is critical. Treatment with a combination of angiotensin-converting enzyme (ACE) inhibitors and loop diuretics increase cardiac output, thereby decreasing the neurohumoral mediators that impair water excretion. Loop diuretics also diminish the action of AVP on the collecting tubules. Thiazides should be avoided because they impair urinary dilution and may worsen hyponatremia. V_2 antagonists increase serum [Na^+] in patients with heart failure, and correction of serum [Na^+] is associated with better long-term outcomes.[35] However, in the much larger randomized controlled EVEREST trial in patients with decompensated heart failure, tolvaptan did not affect long-term clinical outcomes, but some improvement in secondary outcomes, such as urine output, dyspnea, and edema, were noted. In principle, a vaptan with V_1 antagonist activity could have additional benefit in the CHF patient, but this remains unproven.

Cirrhosis. In patients with cirrhosis, water and sodium restriction is the mainstay of therapy. Loop diuretics increase C_{water}. V_2 antagonists increase water excretion and increase serum [Na^+].[35] The response to vaptans in cirrhosis is more attenuated than in patients with SIADH or CHF, which suggests that vasopressin-independent mechanisms may also contribute to the hyponatremia. The administration of V_2 antagonists to patients with liver failure is not associated with decrements in blood pressure. Combined V_1 and V_2 antagonists (e.g., conivaptan) should not be used in these patients, and in view of the potential liver toxicity, the use of tolvaptan should probably be limited to management of serum hyponatremia before liver transplant.[35]

HYPERNATREMIC DISORDERS

Hypernatremia is defined as serum [Na^+] greater than 145 mmol/L and reflects serum hyperosmolarity. The renal concentrating mechanism provides the first defense mechanism against water depletion and hyperosmolarity. The components of the normal concentrating mechanism are shown in Fig. 9.9. Disorders of urine concentration may result from decreased delivery of solute (with decreasing GFR) or the inability to generate interstitial hypertonicity because of decreased Na^+ and Cl^- reabsorption in the ascending limb of Henle's loop (loop diuretics), decreased medullary urea accumulation (poor dietary intake), or alterations in medullary blood flow. Hypernatremia may also result from failure to release or respond to vasopressin. Thirst is the first and most important defense mechanism in preventing hypernatremia.

Etiology and Classification of Hypernatremia

Patients with hypernatremia fall into three broad categories based on volume status.[13] A diagnostic algorithm is helpful in the evaluation of these patients (Fig. 9.10).

Hypovolemia: Hypernatremia Associated With Low Total Body Sodium

Patients with hypovolemic hypernatremia sustain losses of both Na^+ and water but with a relatively greater loss of water. On physical examination, there are signs of hypovolemia, including orthostatic hypotension, tachycardia, flat neck veins, poor skin turgor, and altered mental status. Patients generally have hypotonic water loss from the kidneys or the GI tract; in the GI tract, the urine [Na^+] will be low.

Hypervolemia: Hypernatremia Associated With Increased Total Body Sodium

Hypernatremia with increased total body Na^+ is the least common form of hypernatremia. It results from the administration of hypertonic solutions such as 3% NaCl and $NaHCO_3$ for the treatment of metabolic acidosis, hyperkalemia, and cardiorespiratory arrest. It may also result from inadvertent dialysis against a dialysate with a high Na^+ concentration or from consumption of salt tablets. Therapeutic hypernatremia is also becoming common as hypertonic saline solutions have emerged as an alternative to mannitol for treatment of increased intracranial pressure.[25] Hypernatremia is also increasingly recognized in hypoalbuminemic hospitalized patients with kidney failure who are edematous and unable to concentrate their urine.

Euvolemia: Hypernatremia Associated With Normal Body Sodium

Most patients with hypernatremia secondary to water loss appear euvolemic with normal total body Na^+ because loss of water without Na^+ does not lead to overt volume contraction unless severe. Water loss need not result in hypernatremia unless unaccompanied by water intake. Because hypodipsia is uncommon, hypernatremia usually develops only in those who have no access to water and in very young children and old persons, who may have an altered perception of thirst. Extrarenal water loss occurs from the skin and respiratory tract in febrile or other hypermetabolic states and is associated with a high urine osmolality because the osmoreceptor-AVP renal response is intact. The urine Na^+ concentration varies with intake. Renal water loss leading to euvolemic hypernatremia results either from a defect in AVP production or release (central diabetes insipidus) or from a failure of the collecting duct to respond to the hormone (nephrogenic diabetes insipidus). Defense against the development of hyperosmolality requires the appropriate stimulation of thirst and the patient's ability to respond by drinking water.

Polyuric disorders can result from either an increase in C_{osm} or an increase in C_{water}. An increase in C_{osm} occurs with loop diuretic use, renal salt wasting, excess salt ingestion, vomiting (bicarbonaturia), alkali administration, and administration of mannitol (as a diuretic,

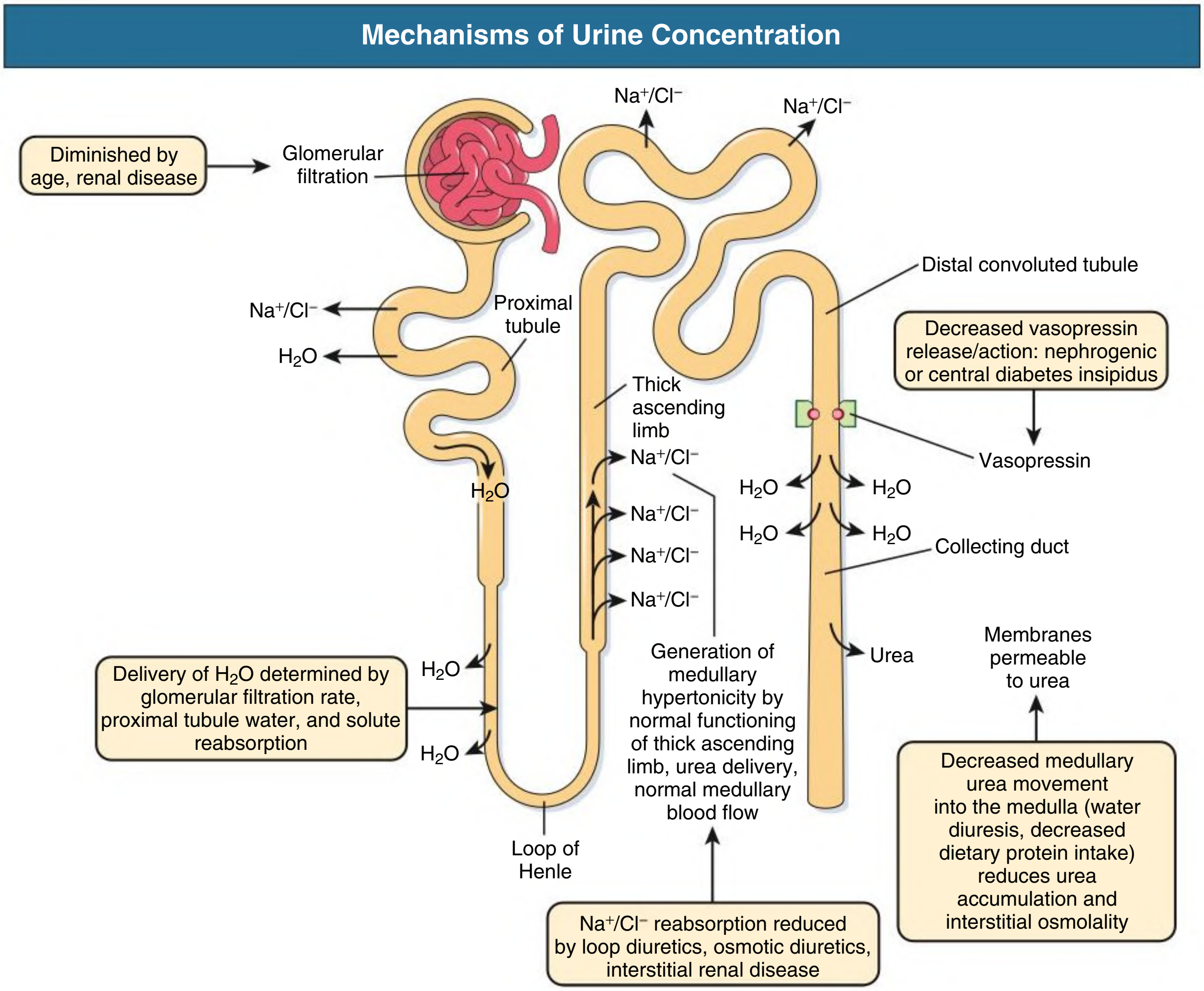

Fig. 9.9 **Urine-Concentrating Mechanisms.** Determinants of normal urine concentration and disorders causing hypernatremia. (Modified from Cogan MC. Normal water homeostasis. In: Cogan MC, ed. *Fluid and Electrolytes*. Lange; 1991:98–106.)

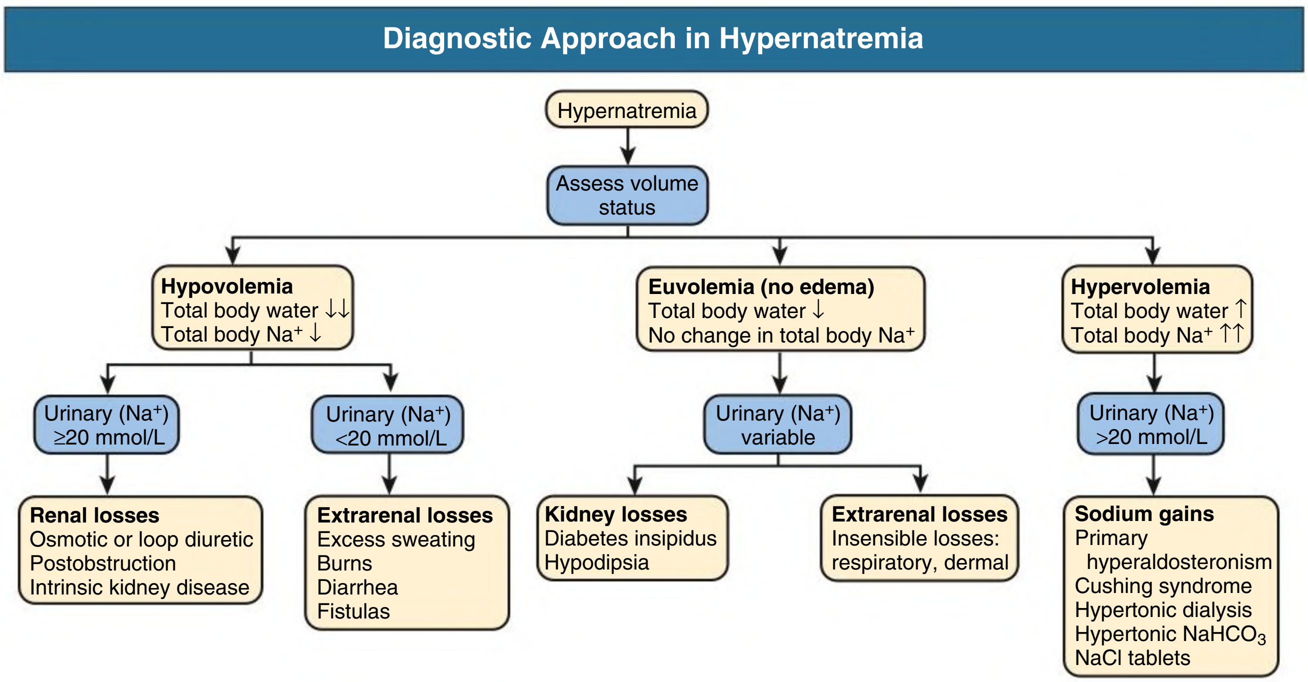

Fig. 9.10 Algorithm for diagnostic assessment of the patient with hypernatremia. (Modified from Halterman R, Berl T. Therapy of dysnatremic disorders. In: Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. Saunders; 1999:257–269.)

TABLE 9.6 Interpretation of Water Deprivation Test

Condition	Urinary Osmolality With Water Deprivation (mOsm/kg H_2O)	Serum Vasopressin After Dehydration (pg/mL)	Increase in Urinary Osmolality With Exogenous Vasopressin or Desmopressin
Normal	>800	>2	Little or no increase
Complete central diabetes insipidus	<300	Undetectable	Substantially increased
Partial central diabetes insipidus	300–800	<1.5	Increase of >10% of urinary osmolality after water deprivation
Nephrogenic diabetes insipidus	<300–500	>5	Little or no increase
Primary polydipsia	>500	<5	Little or no increase

From Lanese D, Teitelbaum I. Hypernatremia. In: Jacobson HR, Striker GE, Klahr S, eds. *The Principals and Practice of Nephrology*. 2nd ed. Mosby; 1995:893–898.

for bladder lavage, or for the treatment of cerebral edema). An increase in C_{water} occurs with excess ingestion of water (psychogenic polydipsia) or in abnormalities of the renal concentrating mechanism (DI).

Diabetes Insipidus

DI is characterized by polyuria and polydipsia and is caused by defects in AVP action. Patients with central and nephrogenic DI and primary polydipsia present with polyuria and polydipsia. These entities can be differentiated by clinical evaluation, with measurements of AVP levels and response to a water deprivation test, followed by AVP administration (Table 9.6).[36]

WATER DEPRIVATION TEST

Test procedure: Water intake is restricted until the patient loses 3% to 5% of body weight or until three consecutive hourly determinations of urinary osmolality are within 10% of each other. (Caution must be exercised to ensure patient does not become excessively dehydrated.) Aqueous vasopressin is given as 5 units subcutaneously, and urinary osmolality is measured after 60 minutes. Expected responses are outlined in Table 9.6.

BOX 9.2 Causes of Central Diabetes Insipidus

Congenital Causes
- Autosomal dominant
- Autosomal recessive

Acquired Causes
- *Posttraumatic*
- *Iatrogenic (postsurgical)*
- *Tumors (metastatic from breast, craniopharyngioma, pinealoma)*
- *Histiocytosis*
- Granuloma (tuberculosis, sarcoid)
- Aneurysm
- Meningitis
- Encephalitis
- Guillain-Barré syndrome
- *Drugs*
- Idiopathic

Entries in italics are the most common causes.

Central diabetes insipidus

Clinical features. Central DI usually has an abrupt onset. Patients have a constant need to drink, have a predilection for cold water, and typically have nocturia. By contrast, the compulsive water drinker may give a vague history of the onset and has large variations in water intake and urine output. Nocturia is unusual in compulsive water drinkers. A serum osmolality of more than 295 mOsm/kg H_2O suggests central DI, and less than 270 mOsm/kg H_2O suggests compulsive water drinking.

Causes. Central DI is caused by infection, tumors, granuloma, and trauma affecting the CNS in 50% of patients; in the other 50% it is idiopathic (Box 9.2). In a survey of 79 children and young adults, central DI was idiopathic in half the patients. The other half had tumors or Langerhans cell histiocytosis; these patients had an 80% risk for development of anterior pituitary hormone deficiency compared with the patients with idiopathic disease.[5]

Autosomal dominant DI is caused by point mutations in a precursor gene for vasopressin that cause "misfolding" of the provasopressin peptide, preventing its release from the hypothalamic and posterior pituitary neurons.[5] Patients present with a mild polyuria and polydipsia in the first year of life. These children have normal physical and mental development. There is a rare autosomal recessive central DI associated with diabetes mellitus, optic atrophy, and deafness (Wolfram syndrome).[37] DI is usually partial with gradual onset in Wolfram syndrome. The defect resulting in Wolfram syndrome is linked to the *WFS1* gene, which encodes an endoplasmic reticulum membrane-embedded protein called wolframin that is expressed in pancreatic beta cells and neurons.

A rare clinical entity involving the combination of central DI and deficient thirst has been reported in approximately 100 patients. When vasopressin secretion and thirst are both impaired, affected patients are vulnerable to recurrent episodes of hypernatremia. Formerly called essential hypernatremia, the disorder is now called central DI with deficient thirst, or *adipsic* DI.[38]

Differential diagnosis. Under basal conditions, AVP levels are unhelpful because there is a significant overlap among the polyuric disorders. Measurement of circulating AVP or copeptin levels after a water deprivation test is more useful (see Table 9.6).

Treatment. Central DI is treated with hormone replacement or pharmacologic agents (Table 9.7). In acute settings, when renal water losses are extensive, DDAVP is the treatment of choice. For chronic central DI, desmopressin acetate is the agent of choice. It has a long half-life and none of the significant vasoconstrictive effects of aqueous AVP. DDAVP is administered at the dose of 10 to 20 μg intranasally every 12 to 24 hours. It is tolerated well, safe to use in pregnancy, and resistant to degradation by circulating vasopressinase. Oral DDAVP (0.1–0.8 mg every 12 hours) is available as second-line therapy. In patients with partial DI, in addition to DDAVP itself, agents that potentiate the release of AVP may be used, including chlorpropamide, clofibrate, and carbamazepine.

Congenital nephrogenic diabetes insipidus. Inherited forms of DI are caused by mutations in genes for V_2-receptors or AQP2.[5] These entities are discussed further in Chapter 49. Urine volumes are typically very high, and there is a risk for severe hypernatremia if patients do not have free access to water. Thus, patients must drink enough water to match urine output and prevent dehydration. Therapy for congenital nephrogenic DI is only partially effective and includes a thiazide diuretic, a very low salt diet, and indomethacin. Sildenafil improved urine concentration in a patient with congenital nephrogenic DI.[39] Simvastatin induces an increase in urine AQP2 and osmolality in hypercholesterolemic patients, suggesting that it may be useful in congenital nephrogenic DI.[39] Metformin results in a sustained increase in urine osmolality in rodent models of nephrogenic diabetes insipidus, but it has not been tested in patients.[39]

Acquired nephrogenic diabetes insipidus. Acquired nephrogenic DI is more common than congenital nephrogenic DI but rarely as severe. In patients with acquired nephrogenic DI, the ability to elaborate a maximal concentration of urine is impaired, but urine-concentrating mechanisms are partially preserved. For this reason, urine volumes are less than 3 to 4 L/day, which contrasts with the much higher volumes seen in patients with congenital or central DI or compulsive water drinking. Table 9.8 outlines the causes and mechanisms of acquired nephrogenic DI.

Chronic kidney disease. A defect in urine-concentrating ability may develop in patients with CKD of any etiology, but this defect is most prominent in tubulointerstitial diseases, particularly medullary cystic disease. (A complete discussion on the mechanisms of abnormalities in urine concentration and dilution in CKD can be found in Berl and Combs[13]). Disruption of inner medullary structures and diminished medullary concentration are thought to play a role; alterations in V_2 receptor and AQP2 expression also contribute (see Fig. 9.6). To achieve daily osmolar clearance, patients should be advised to maintain a fluid intake that matches their urine volume.

Electrolyte disorders. Hypokalemia causes a reversible abnormality in urine-concentrating ability. Hypokalemia stimulates water intake and reduces interstitial tonicity, which relates to the decreased Na^+-Cl^- reabsorption in the TAL. Hypokalemia resulting from diarrhea, chronic diuretic use, and primary aldosteronism also decreases intracellular cyclic adenosine monophosphate accumulation and causes a reduction in AVP-sensitive AQP2 expression (see Fig. 9.6).

Hypercalcemia also impairs urine-concentrating ability, resulting in mild polydipsia. The pathophysiologic mechanism is multifactorial and includes a reduction in medullary interstitial tonicity caused by decreased vasopressin-stimulated adenylyl cyclase in the TAL and a defect in adenylyl cyclase activity with decreased AQP2 expression in the collecting duct, mediated by the calcium-sensing receptor.[7]

Pharmacologic agents. Lithium is the most common cause of nephrogenic DI, occurring in up to 50% of patients receiving long-term lithium therapy. Lithium causes downregulation of AQP2 in the collecting duct; experimentally, it also increases cyclooxygenase-2 (COX-2) expression and urinary prostaglandins, which may contribute to the polyuria.[7] The concentrating defect of lithium may persist even when the drug is discontinued. The epithelial sodium channel (ENaC) is the entrance pathway for lithium into collecting duct principal cells. Amiloride inhibits lithium uptake through ENaC and has been used clinically to treat nephrogenic DI caused by lithium. Aldosterone administration dramatically increased urine production in experimental nephrogenic DI caused by lithium (an effect associated with decreased expression of AQP2 on luminal membranes of collecting duct), whereas administration of the mineralocorticoid receptor blocker spironolactone decreased urine output and increased AQP2 expression.[7] Clopidogrel, P2Y12-R purinergic receptor antagonist, ameliorates lithium-induced NDI in mice by increasing water and sodium reabsorption in the collecting duct.[39] It is not yet known if spironolactone or clopidogrel will be a useful treatment for lithium-induced nephrogenic DI in patients.

Other drugs impairing urine-concentrating ability include amphotericin, foscarnet, and demeclocycline, which reduce renal medullary adenylyl cyclase activity, thereby decreasing the effect of AVP on the collecting ducts.

Sickle cell anemia. Patients with sickle cell disease and trait often have a urine-concentrating defect. In the hypertonic medullary interstitium, the "sickled" red cells cause occlusion of the vasa recta and papillary damage. Although initially reversible, medullary infarcts

TABLE 9.7 Treatment of Central Diabetes Insipidus

Disease	Drug	Dose	Interval (hr)
Complete central diabetes insipidus	Desmopressin (DDAVP)	10–20 µg intranasally	12–24
	Desmopressin (DDAVP)	0.1–0.8 mg orally	Every 12
Partial central diabetes insipidus	Desmopressin (DDAVP)	10–20 µg intranasally	12–24
	Aqueous vasopressin	5–10 U subcutaneously	4–6
	Chlorpropamide	250–500 mg	24
	Clofibrate	500 mg	6 or 8
	Carbamazepine	400–600 mg	24

TABLE 9.8 Acquired Nephrogenic Diabetes Insipidus: Causes and Mechanisms

Disease State	Defect in Medullary Interstitial Tonicity	Defect in cAMP Generation	Downregulation of Aquaporin 2	Other
Chronic kidney disease	Yes	Yes	Yes	Downregulation of V_2 receptor message
Hypokalemia	Yes	Yes	Yes	—
Hypercalcemia	Yes	Yes	—	—
Sickle cell disease	Yes	—	—	—
Protein malnutrition	Yes	—	Yes	—
Demeclocycline therapy	—	Yes	—	—
Lithium therapy	—	Yes	Yes	—
Pregnancy	—	—	—	Placental secretion of vasopressinase

cAMP, Cyclic adenosine monophosphate.

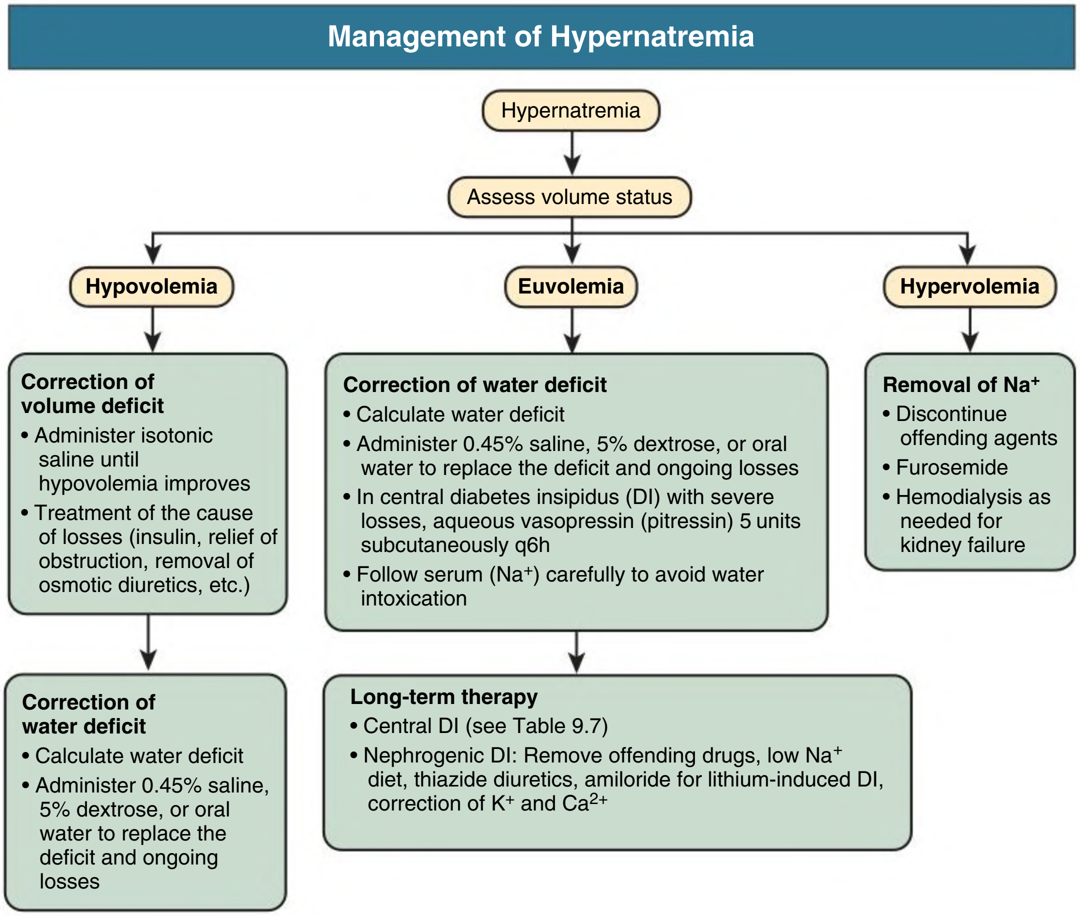

Fig. 9.11 Algorithm for management of the patient with hypernatremia. (Modified from Thurman JM, Berl T. Therapy of dysnatremic disorders. In: Wilcox CS, ed. *Therapy in Nephrology and Hypertension*. 3rd ed. Saunders; 2008:337–352.)

occur with sickle cell disease in the second to third decade of life, and even with sickle cell trait by the fourth to fifth decade, and the concentrating defects become irreversible.

Dietary abnormalities. Extensive water intake or a marked decrease in salt and protein intake leads to impairment of maximal urine-concentrating ability through a reduction in medullary interstitial tonicity. On a low-protein diet with excessive water intake, there is a decrease in vasopressin-stimulated osmotic water permeability that is reversed with feeding.

Gestational diabetes insipidus. In gestational DI, there is an increase in circulating vasopressinase, which is produced by the placenta. Patients are typically unresponsive to AVP but respond to desmopressin, which is resistant to vasopressinase.

Clinical Manifestations of Hypernatremia

Certain patients are at increased risk for development of severe hypernatremia (Box 9.3). Signs and symptoms mostly relate to the CNS and include altered mental status, lethargy, irritability, restlessness, seizures (usually in children), muscle twitching, hyperreflexia, and spasticity. Fever, nausea or vomiting, labored breathing, and intense thirst can also occur. In children, mortality of acute hypernatremia ranges from 10% to 70%; as many as two-thirds of survivors have neurologic sequelae. In contrast, mortality in patients with chronic hypernatremia is 10%.

In adults, serum [Na^+] greater than 160 mmol/L is associated with 75% mortality, although this may reflect associated comorbidities rather than hypernatremia. Chronic hypernatremia is independently associated with higher mortality in patients with CKD.[40] Patients presenting to the intensive care unit with a serum [Na^+] greater than 155 mmol/L have a significantly increased risk of mortality, with an odds ratio of 3.64.[41] Thus, both acute and chronic hypernatremia are associated with an increased risk of mortality.

BOX 9.3 Patient Groups at Risk for Development of Severe Hypernatremia

- Elderly patients
- Infants
- Hospitalized patients
  - Hypertonic infusions
  - Tube feedings
  - Osmotic diuretics
  - Lactulose
  - Mechanical ventilation
- High-risk patient groups
  - Altered mental status
  - Uncontrolled diabetes mellitus
- Underlying polyuric disorders

From Thurman JM, Berl T. Therapy of dysnatremic disorders. In: Wilcox CS, ed. *Therapy in Nephrology and Hypertension.* 3rd ed. Philadelphia: Elsevier; 2008:337–352.

Treatment of Hypernatremia

Hypernatremia occurs in predictable clinical settings, allowing opportunities for prevention. Elderly and hospitalized patients are at high risk because of impaired thirst and inability to access free water independently.[42] Certain clinical situations, such as recovery from AKI, catabolic states, therapy with hypertonic solutions, uncontrolled diabetes, and burns, should prompt close attention to serum sodium concentration and increased administration of free water.

Hypernatremia always reflects a hyperosmolar state. The primary goal in the treatment of these patients is the restoration of serum tonicity. Fig. 9.11 outlines specific management options.[12] Because restoration of volume takes precedence over restoration of tonicity, in hypovolemic hypernatremic patients, sodium-containing solutions should be used until euvolemia is achieved. Thereafter, dextrose in water or oral water intake should be given to decrease serum sodium concentration.

The rapidity with which hypernatremia should be corrected is controversial. Some animal studies and case series in pediatric patients suggest that a correction rate of more than 0.5 mmol/L/hr in [Na^+] can cause seizures. Cerebral edema also can be caused by rapid correction of hypernatremia by the net movement of water into the brain. Most clinicians believe that even in adults, correction should be achieved during 48 hours at a rate no greater than 2 mmol/L/hr, but the current maximum rate for correction in adults has not been definitively established because there are no reports of cerebral edema with rapid correction in this population. One study in a population of chronically hypernatremic patients showed that there was no difference in 30-day mortality between rapid correction (>12 mmol/L/day) versus less than 12 mmol/L/day, so a more rapid correction could potentially be a safe option.[43]

SELF-ASSESSMENT QUESTIONS

1. A 78-year-old woman sustained a stroke and is receiving enteral nutrition by nasoduodenal tube at a nursing home. She is transferred to the hospital with altered mental status. Nursing home staff reported that she had plentiful urine output 2 days before transfer. Laboratory data on hospital arrival reveal serum sodium of 165 mmol/L, blood urea nitrogen (BUN) 60 mg/dL, and creatinine 1.6 mg/dL; all were normal 6 weeks earlier. Additional laboratory data include serum osmolality of 341 mOsm/kg and urine osmolality 690 mOsm/kg, urine sodium 7 mmol/L, and urine potassium 32 mmol/L. The most likely explanation for this patient's polyuria and hypernatremia and the most appropriate confirmatory test are:
   A. Nephrogenic diabetes insipidus and measurement of ADH.
   B. Central diabetes insipidus and measurement of ADH.
   C. Lithium toxicity and measurement of serum lithium level.
   D. Increased solute load and measurement of electrolyte-free water clearance.
2. A 71-year-old woman with a history of coronary artery disease and mild hypertension is seen in the clinic. She is free of symptoms. Her medications include lisinopril (20 mg/day), occasional zolpidem tartrate (Ambien), and multivitamins. Her blood pressure is 140/95 mm Hg, weight 62 kg, and skin turgor normal. Examination reveals no evidence of edema or ascites. Laboratory results are as follows:
   - Serum creatinine: 0.9 mg/dL.
   - Sodium: 120 mmol/L.
   - Potassium: 3.9 mmol/L.
   - Chloride: 95 mmol/L.
   - HCO_3^-: 22 mmol/L.
   - U_{osm}: 686 mOsm/kg.
   - U_{Na}: 127 mmol/L.
   - Normal thyroid and adrenal function.
   - Chest radiograph unremarkable.

   Which of the following statements explains this patient's status?
   A. Patient is unlikely to have improved water excretion because she is not taking a thiazide diuretic.
   B. Patient probably has idiopathic SIADH.
   C. The hyponatremia is probably a consequence of poor solute intake.
   D. Patient's hyponatremia is caused by age-related decrements in vasopressin metabolism.
3. A 74-year-old man with a history of heart failure is admitted with increasing shortness of breath. He is treated with lisinopril 20 mg/day, hydrochlorothiazide (HCTZ) 50 mg/day, and digoxin 0.25 mg/day. Examination reveals blood pressure of 145/90 mm Hg, pulse rate 88/min, respiratory rate 24/min, and O_2 saturation 90% on 2 L. His electrolytes are normal. The patient is prescribed a low-sodium restricted diet, furosemide 40 mg twice daily, and fluid restriction. In the next 24 hours, he excretes 2.5 L of urine. Laboratory results are as follows:
   - Serum creatinine: 1.8 mg/dL.
   - BUN: 25 mg/dL.
   - Sodium: 147 mmol/L.
   - Potassium: 3.7 mmol/L.
   - HCO_3^-: 26 mmol/L.
   - Chloride: 120 mmol/L.
   - U_{osm}: 392 mOsm/kg.
   - U_{Na}: 59 mmol/L.
   - U_K: 32 mmol/L.

   Which of the following treatment regimens is most likely to prevent worsening of this patient's hypernatremia?
   A. 1 mL of half-normal per milliliter of urine.
   B. 0.5 mL of 5% dextrose in water per milliliter of urine.
   C. 0.5 mL of normal saline per milliliter of urine.
   D. 0.5 mL of normal saline per milliliter of urine.

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